Reinforcement Learning for Optimizing Marketing Discount Strategies

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***Abstract*— This research aims to apply Reinforcement Learning (RL) to the marketing department by using marketing strategies that offer appropriate discounts based on the income classification of customers. The study focuses on using Q-learning. The project presents an intelligent agent capable of making appropriate decisions based on each customer's income, which is classified into three levels: low, medium, and high. The type of discount is chosen according to each customer's income level, with discounts classified into three types: 10%, 20%, and no discount, in addition to holiday-specific discounts of 30%. The agent receives rewards or penalties based on the final results, and it continuously learns from the outcomes it achieves, updating its strategy using the Q-learning algorithm.**

The results from the graphs showed that customers classified under the high-income level respond more to discounts that occur during holidays and events, unlike customers classified under the low-income level, who tend to purchase products even without promotional offers. Ultimately, it was found that the agent is highly capable of making very appropriate marketing decisions, as it improved sales while simultaneously maintaining profits. This indicates that making marketing decisions through the agent is much better than making marketing decisions based on traditional methods. Additionally, this project provided a scientific platform capable of development to improve marketing decision-making processes, increase sales, and maintain profits to a greater extent, and this development is achieved through artificial intelligence.

Keywords*— *Reinforcement Learning, Discount Strategy, Q-Learning, Customer Segmentation, Sales Optimization

I. Introduction

Choosing the right discount for each type of customer can really help a business increase its sales without reducing profits. That's where reinforcement learning (RL) can help. It's a type of artificial intelligence that learns from experience to make better decisions over time.

In our project, we explored how RL can be used to determine the best discount based on customer income level. We segmented customers into three groups: low, medium, and high income. Then, we tested different discount options: no discount, 10%, 20%, and a special offer during major events like holidays. Using Python, the system learns which discount works best for each group based on whether the customer decides to make a purchase or not.

The goal is to help businesses offer smarter promotions that match customer needs and improve overall performance.

II. Literature review

Reinforcement learning (RL) has recently garnered significant attention, becoming a widely adopted approach in the model of intelligent decision-making systems across a vast array of fields. As Diane and Neve (2008) explained in their study of the computational, neural, and biological foundations of reinforcement learning, the researchers applied these foundations to simulate how humans (the brain) learn from experiences, as they learn based on their experiences and surrounding environment. The researchers confirmed that reinforcement learning learns from prediction errors of rewards, reflecting the dopaminergic process in the brain, which is exactly what happens in the human brain when learning from experiences. This study also included a distinction between model-free and model-based learning, and each one's role in behavioral adaptation. Model-free learning learns through experience and is less flexible compared to model-based learning, while model-based learning learns through pre-planned inputs. The study confirmed that reinforcement learning supports sequential decision-making through actions based on environmental feedback. This study provided results related to the fundamentals of applying algorithms present in reinforcement learning. These different algorithms within reinforcement learning are applied in various fields such as marketing, where the agent can study the different environments of each customer and make sound decisions in offering suitable promotions, thereby using its intelligence to gain new customers and increase profits.

Chen et al. (2020) conducted a comprehensive study on decision-making in energy systems such as renewable energy, modern energy, and distributed energy resources based on reinforcement learning (RL). The study focused heavily on Deep Learning and Q-learning, as with the increasing amount of data, the data becomes highly complex, and decision-making becomes extremely difficult. It is always recommended to use Deep Learning in these cases. However, when working on a small project that does not require a lot of data, it is recommended to use Q-learning. Researchers confirmed that using Deep Learning helps in facing challenges related to large data and making the right decisions through it. The study also delved into various aspects, including algorithms and methods based on value and policy, such as Q-learning. It discussed their applications in three main areas: frequency regulation to maintain network stability, voltage control to ensure power quality, and energy management for efficient power distribution. In the end, the results showed that reinforcement learning is capable of excelling with its new methods in decision-making instead of the traditional and old

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methods based on uncertain dynamic environments. This study helps in applying reinforcement learning in various fields such as marketing, where decision-making in the face of uncertainty and personalization is of utmost importance. It also highlights the necessity of using environments like Deep Learning that support complex and difficult data to ensure correct decision-making. Additionally, in marketing, the store must evolve in the future and increase the number of customers more and more, requiring environments that suit the work and make decision-making precise and simple.

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Dulac-Arnold et al. (2019) examined the barriers that limit the deployment of reinforcement learning in real-world applications, emphasizing the challenges that differ from controlled experimental settings. The study pointed out issues such as limited data, partial observability, high-dimensional action spaces, and the need for safety, interpretability, and real-time responses. These challenges often prevent reinforcement learning algorithms from being directly applied in practical scenarios. To address these concerns, the researchers introduced the "realworldrl-suite," a benchmark framework that simulates real-world conditions to test and improve RL algorithms. Their work underscores the importance of robust and flexible RL methods that can operate under uncertainty, making it highly relevant for dynamic domains such as marketing, where decisions must be personalized, adaptive, and interpretable.

Nguyen et al. (2020) presented a comprehensive review of deep reinforcement learning (DRL) within multi-agent systems (MAS), highlighting its growing significance in dynamic and decentralized environments. The study addressed key challenges in MAS applications, such as non-stationarity, partial observability, and scalability, which complicate learning and coordination between agents. To overcome these barriers, the authors surveyed advanced solutions including centralized training with decentralized execution, communication protocols between agents, and hierarchical reinforcement learning. The paper also outlined practical applications of DRL in areas like autonomous vehicles, robotics, and network management, where coordination between multiple intelligent agents is crucial. These findings emphasize the flexibility and scalability of reinforcement learning, particularly in environments requiring collective intelligence, and support its relevance in complex decision-making contexts such as healthcare, marketing, and energy systems.

Sun et al. (2022) explored the use of deep reinforcement learning (DRL) to optimize market making strategies in financial trading. The study proposed a DRL-based framework capable of learning optimal bid and ask quotes while balancing profit and inventory risk in real-time. Unlike traditional rule-based or stochastic models, the DRL model adapts to dynamic market conditions and learns through interaction with the environment. The authors employed a Double DQN algorithm to improve learning stability and prevent overestimation of action values. Experimental results demonstrated that the DRL-based strategy significantly outperformed baseline methods in terms of profitability and risk control. This study underscores the practical value of reinforcement learning in financial decision-making, particularly in fast-paced and uncertain markets where adaptability is critical.

III. Problem Statement Definition

In today's marketing world, one of the major challenges companies faces is how to decide the most appropriate discount for different types of customers. Offering the same discount to everyone can lead to wasted resources and missed opportunities. For example, giving a high discount to high-income customers who would have purchased anyway reduces profit, while offering a small discount to low-income customers might not be enough to convince them to buy.

This highlights the need for an intelligent system capable of learning from customer behavior and recommending the most suitable discount based on individual customer profiles. To address this, we apply concepts from Artificial Intelligence (AI), particularly Reinforcement Learning (RL).

In our project, we use a specific algorithm called Q-learning, one of the most well-known techniques in RL. The idea is to train an "agent" that interacts with the environment (customers), learns from their responses (purchase or no purchase), and improves its decision-making over time.

Customers are divided into three income levels: low, medium, and high income. To determine the most effective marketing strategy for each group, we test several discount options, including no discount, a 10% discount, a 20% discount, and a Special 30% discount typically used during holidays.

The agent tries different discounts for each customer type and observes the results. The agent receives a positive bonus (+1), if the customer makes a purchase. Over time, the agent learns which discount is best for each income group.

Q-learning works by updating a "Q-table" that stores the value of taking an action in a particular state. The agent updates the table using the following equation:

$$Q(s,a) = Q(s,a) + \propto [R + \gamma \, maxQ(s',a') - Q(s,a)]$$

Where $Q(s,a)$ represents the current estimated value of taking action a in state s, $\propto$ is the learning rate which determines how quickly the agent updates its knowledge based on new experiences, R is the reward received after performing the action, γ is the discount factor reflecting how much the agent values future rewards compared to immediate ones, and $maxQ(s',a')$ is the best reward the agent thinks it can get from the next step by picking the right action. Using this formula iteratively, the agent gradually learns which discount strategy is most effective for each customer income group by maximizing the expected long-term reward.

IV. Results and Discussions

The Reinforcement Learning (RL) process was designed to allow an agent to learn the optimal discount strategy for different customer income levels. By simulating multiple interactions with the environment, the agent selected actions (discount offers), observed the outcomes, received rewards, and updated its knowledge accordingly through the Q-learning algorithm.

After sufficient training, the agent's Q-table reflected learned preferences per income group, helping identify the best discount strategy for each. This learning method works like common reinforcement learning methods, where the agent tries to get as many rewards as possible over time.

To apply reinforcement learning in our marketing discount strategy, we structured the problem based on the core elements of the RL frameworks: the agent, environment, states, actions, and rewards. This interaction underscores the importance of having an intelligent system that can continuously learn from customer responses and dynamically recommend the most appropriate discount based on each customer's profile. The following diagram illustrates this learning process and how the agent gradually improves its decision-making over time (look to **figure 1**).

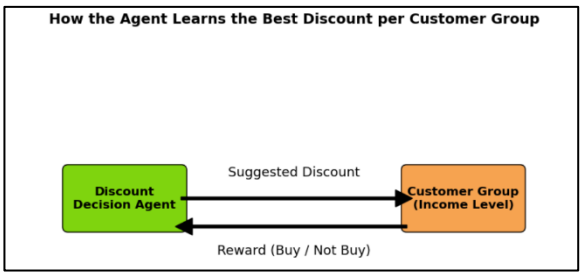

Figure 1 *Reinforcement Learning Process*

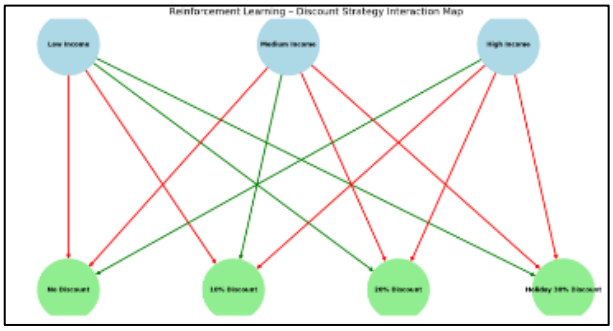

Figure 2 *Reinforcement Learning -Discount Strategy Interaction Map*

Figure 2 *This figure illustrates the interaction between different income levels (Low, Medium, High) and various discount offers (No Discount, 10%, 20%, 30% Holiday Discount). Arrows connect income groups to discount actions, where green arrows indicate positive outcomes (e.g., purchase behavior or high reward), and red arrows indicate negative outcomes (e.g., no purchase or low reward). This visual highlights how the reinforcement learning agent learned which discount strategies are most effective for each customer segment.*

```
Q-table after training:
[[3.18097698 1.79070354 5.38545398 3.33504732]
 [4.57553201 3.20681031 1.82904329 2.89847049]
 [5.60972436 2.42844858 2.81643892 3.14179078]]
```

Figure 3 *Q-table after Training*

The **Q-table** shows what the agent has learned about which actions (discount options) are best for each income group. Each value represents how effective a discount is for that customer type. A higher Q-value indicates a more beneficial strategy. These values were used to determine the best discount per group.

	No Discount	10% Discount	20% Discount	30% Holiday Discount
Low Income	0.149281	0.193813	0.930223	0.244712
Medium Income	0.658146	0.937581	0.697466	0.641657
High Income	0.786183	0.842098	0.114562	0.760559

Figure 4 *Q-table as Formatted Table*

Here, the **Q-table (figure 4)** is presented in a clean, readable tabular format. Each row corresponds to an income group, and each column shows the Q-value for a particular discount. This presentation facilitates direct comparisons.

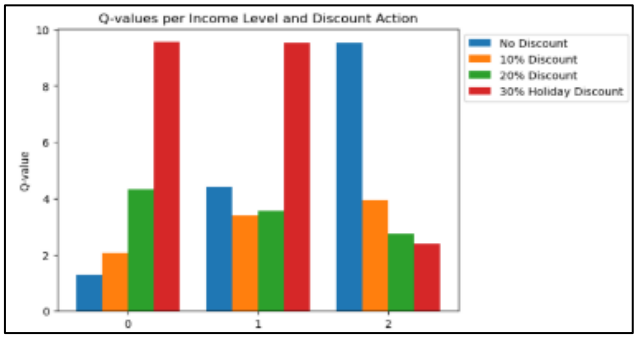

Figure 5 *Grouped Bar Chart – Q-values per Income Level and Discount Action*

This grouped bar chart (**figure 5**) presents a clear visual representation of the Q-values obtained for each discount strategy across different income levels. The x-axis displays the three income categories (low, medium, and high), while the y-axis shows the Q-values, which indicate the expected long-term reward for each action as learned by the reinforcement learning agent.

Each income group is associated with four bars, each corresponding to a specific discount offer: No Discount, 10% Discount, 20% Discount, and 30% Holiday Discount. By comparing the heights of the bars, we can observe how effective each discount strategy is for each customer segment.

The chart shows patterns and choices among different groups. For instance, lower-income customers might show a strong response to larger discounts like the 30% holiday offer, indicated by a higher Q-value for that action. On the other hand, higher-income customers may not respond as strongly to discounts, which can be seen if the "No Discount" or lower discount bars are relatively higher.

This visualization allows decision-makers to quickly assess which discount strategies yield the highest potential rewards for each income level. It complements the numerical Q-table by providing an intuitive view of the model's learned preferences, supporting data-driven marketing strategies that align with customer behaviors.

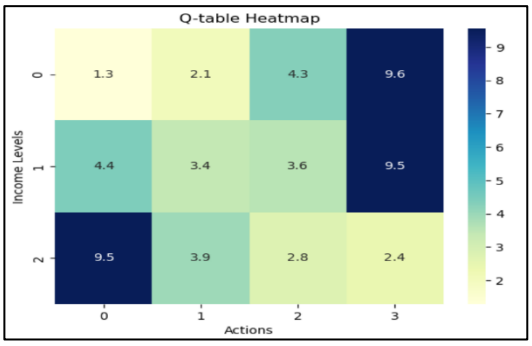

Figure 6 *Q-table Heatmap*

This heatmap (**figure 6**) shows the Q-table using colors to represent how big or small the Q-values are. Rows correspond to income levels (0 = Low, 1 = Medium, 2 = High), and columns represent discount actions (0 = No Discount, 1 = 10% Discount, 2 = 20% Discount, 3 = 30% Holiday Discount).

Darker shades (e.g., deep blue) indicate higher Q-values, suggesting more effective discount strategies for that income group. Lighter shades (e.g., yellow) represent lower Q-values, indicating less favorable actions. This format allows for quick identification of the most rewarding discount per income level, making it easier to interpret the agent's learned preferences.

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Low Income → Best offer: No Discount
Medium Income → Best offer: 20% Discount
High Income → Best offer: 30% Holiday Discount
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Figure 7 *Best Action per Income Level*

Figure 7 illustrates the best discount strategy for each income level after the model completed its training. Based on the Q-table values, the agent determined that the most effective offer for low-income customers is no discount, while medium-income customers respond best to a 20% discount. For high-income customers, the 30% holiday discount yielded the highest expected reward. These results suggest that higher-income individuals may be more influenced by larger promotional offers, whereas lower-income customers might convert even without incentives. It's important to note that these optimal actions can vary between training runs due to the randomness in initial Q-values and the agent's exploration behavior during learning.

V. LIMITATIONS

This project has some limitations. First, the Q-learning model is based on a simple setup with only three income levels and four types of discounts. In real life, customers are more different and their behavior is more complex. Also, the agent learns from fake (simulated) data, not real customer data, so the results may not be accurate in real business situations. The rewards we used are based on assumptions and may not match real profits or customer choices. Other important factors, like seasons, customer preferences, or marketing efforts, were not included. Lastly, since the model uses randomness, the results can change each time we run the program unless we fix the random seed.

VI. FUTURE DIRECTIONS

While the current agent has shown great success in making appropriate decisions for the marketing department, specifically in selecting the right offer for each client based on their income level, it can still be improved further to attract more new customers while maintaining profits at the same time.

First, the agent can be developed through the application of Deep Learning to improve marketing strategies. As time passes and the number of customers increases, the data will also increase, making it very difficult to make appropriate decisions quickly and efficiently. The available data will become very large, making it very complex to work with. Therefore, using Deep Learning will greatly help in making appropriate decisions because it is deeper than Q Learning. At the same time, it can handle very large data and can also make appropriate decisions.

Secondly, the agent can be developed by adding new types of discounts to renew the use of discounts, allowing them to attract customers without losing profits. For example, they can offer a customer loyalty card, and with each purchase, the customer gets closer to receiving a suitable discount or gift. In this case, customers will be encouraged to make repeat purchases to reach the goal and receive a discount or gift. Alternatively, they can add gifts to new products instead of offering discounts, as customers will try the new product, which will encourage them to return and buy new products.

Thirdly, the agent's effectiveness can be increased by adding deeper information about customer behavior, such as:

the number of visits by the customer, how long it takes the customer to make a purchase decision, how many times the customer has bought products from our store, and the customer's preferences. This addition will help the agent make appropriate decisions regarding providing the right offer and will enable the agent to anticipate the customer's reaction based on the recorded behaviors.

Fourthly, seasonal discounts can be added that relate to the time and appropriate season for this discount, instead of having a single discount for all seasons and occasions. By adding this feature, the discounts will be more suitable for the time and context, which will increase the number of customers. We can also ensure that fixed and seasonal discounts do not conflict with each other, as when there is a seasonal offer, the customer cannot use the regular offer because it might cause a loss in the store's profit.

Fifthly, customer satisfaction ratings can be added to obtain rewards or penalties, as this addition will make the agent smarter in decision-making because they will strive for customer satisfaction for a new visit instead of just evaluating whether they made a purchase or not. The following figure 3 shows the feature score obtained by random forest with an accuracy of 0.7667. The main difference is that all RF features become significant in the IA effectiveness.

Sixthly, integrating the agent with real electronic platforms, such as Shein. The agent can set appropriate discounts for each customer based on the data available on the site for each customer purchasing from this site. This will help the agent make decisions in different environments, making them more effective and flexible, and they can continuously learn from different environments.

Seventhly, a simple dashboard can be designed for the marketing team, through which the employees in this team can monitor decision-making processes relying on the agent. When any issue arises, the team can easily and quickly adjust these discrepancies to avoid losses. Additionally, it helps in training employees to use technology in real-life scenarios and increases profits through this system.

These future additional improvements will greatly help enhance this agent, as they will increase its decision-making power to support marketing decisions in work environments, and will improve its effectiveness in attracting many customers, increasing sales, maintaining profit, and boosting competitive strength.

VII. Conclusion

This project demonstrated the discovery and application of the role of reinforcement learning (RL), specifically in the Q-learning algorithm. RL was applied to improve decision-making processes in the marketing department to provide each customer with a suitable discount based on their income level. With each customer's reaction, rewards and discounts are calculated for the agent to ensure its effectiveness and improve its decision-making processes over time. Ultimately, the agent succeeded remarkably in the decision-making process, surpassing traditional techniques in marketing decision-making. The results that emerged confirmed that the agent is based on making personalized decisions grounded in various data, which enhances customer satisfaction and increases business profitability. This agent has been prepared in a practical and flexible manner so that it can be used as a flexible and scalable approach in decision-making processes.

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